

Multiplying matrices

Key Points

To multiply two matrices they must conform meaning that for A of order $n \times m$ and B of order $k \times l$ we must have $m=k$

To do matrix multiplication perform the dot product row of matrix A with column of matrix B

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}$$

Multiplying by an identity remains the same and multiplying by a zero makes it zero

Basic Skills

Q1. What is the order of a matrix

Q2. For $A = \begin{pmatrix} 3 & 4 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -1 \\ 4 & 0 \\ 3 & -1 \end{pmatrix}$, $C = \begin{pmatrix} 5 & 3 & 4 \\ 2 & 1 & 2 \\ 3 & 1 & 1 \end{pmatrix}$ state all products which conform

Q3. a) Explain why $\begin{pmatrix} 1 & 5 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 13$

b) Calculate $\begin{pmatrix} 2 & -4 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

c) Hence calculate $\begin{pmatrix} 1 & 5 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

Q4. Calculate $\begin{pmatrix} 1 & 5 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ 2 & 1 \end{pmatrix}$

Competency Skills

Q5. Use the matrices $A = \begin{pmatrix} 5 & 3 \\ 6 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -3 \\ 4 & 2 \end{pmatrix}$ to show that $AB \neq BA$ for all matrices

Q6. For the matrices $A = \begin{pmatrix} 3 & 4 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -1 \\ 4 & 0 \\ 3 & -1 \end{pmatrix}$, $C = \begin{pmatrix} 5 & 3 & 4 \\ 2 & 1 & 2 \\ 3 & 1 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 5 & 2 & -1 \\ 0 & 2 & 0 \\ 3 & 5 & 1 \end{pmatrix}$, $E = \begin{pmatrix} 2 & -1 & 0 \\ 4 & 2 & 4 \end{pmatrix}$ determine the following products, if it is possible to do so

a) AB **b)** BA **c)** CB **d)** DE **e)** DC **f)** EB **g)** BE

Q7. For $M = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$ by finding $M^2, M^3, \dots$ suggest a general formula for M^n

Q8. Calculate $\begin{pmatrix} x & 0 & 4 \\ 2x & 1 & 3 \\ -2 & 5 & 3-x \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x & -1 & 0 \\ 1 & x & 1 \\ 0 & -1 & x \end{pmatrix}$ in terms of x

Advanced Skills

Q9. A matrix is called symmetric if elements are symmetric about the leading diagonal

a) If $\begin{pmatrix} 3 & x \\ 1 & 4 \end{pmatrix}$ is symmetric what is x

b) Given that $M = \begin{pmatrix} -1 & x \\ x & 2 \end{pmatrix} \begin{pmatrix} 3 & 3 \\ -1 & 4 \end{pmatrix}$ is symmetric find x

Q10. The matrix $S = \begin{pmatrix} 4 & 2 & 1 \\ 10 & 2 & 3 \\ 8 & 5 & 4 \end{pmatrix}$ represents the number of sales of chocolate; plain, caramel, toffee (on rows) against the time of day; morning, afternoon, evening (on cols). Find a matrix M such that MS represents the total number of each chocolate sold for the entire day

Extension

Q11. Consider 4 paths A,B,C,D on 4 levels 1,2,3,4 with only one path per level at a time. Find all possible arrangements and matrices which can transform between them

Hint: The initial arrangement A on 1, B on 2, C on 3, D on 4 can be written as the identity